

PAVNI ARORA

B.Tech Information Technology — VIT Vellore

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Summary

2nd-year B.Tech IT student at VIT Vellore (CGPA: 9.58 — 8th Branch Rank) with S grades across all Mathematics courses. Specializing in deep learning and computer vision: GNNs, real-time CV systems, and object detection pipelines. Seeking to deepen expertise in applied ML through rigorous, research-driven environments.

Education

B.Tech in Information Technology CGPA: 9.58

8th Rank in Branch

Vellore Institute of Technology (VIT), Vellore

2024 – Present

Relevant Coursework: Complex Numbers and Linear Algebra, Probability & Statistics, Calculus, Discrete Mathematics, Differential Equation and Transforms. Awarded S Grade (Outstanding) in all Mathematics courses. (highest grade on VIT's grading scale.)

Technical Skills

Programming Languages: C, C++, Java, Python

Web Development: HTML, CSS, JavaScript, ReactJS, JSON

Computer Science Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems

AI & Machine Learning: OpenCV, Convolutional Neural Networks (CNNs), Generative AI Fundamentals, Transformer

Architectures, PyTorch / TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib

Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook

Projects

DeepShield - Deepfake Detection System

Python, PyTorch, PyTorch Geometric, OpenCV, Streamlit, Deep Learning

- Engineered a hybrid ResNet18 + GCN (FuNet-M) architecture and graph-based facial feature modeling.
- Implemented custom graph generation, training, and inference pipelines using PyTorch and PyTorch Geometric.
- Achieved 98.81% accuracy, 98.80% F1-score, and 99.38% ROC-AUC on a balanced deepfake dataset of 416 facial frames (for rapid prototyping.)

Sketch2Code

ReactJS, Tailwind CSS

- Built a canvas-based UI designer in React where developers draw rectangles and instantly get production-ready Tailwind JSX, CSS, and utility classes; eliminating trial-and-error pixel guessing.
- Implemented live SVG canvas with drag-to-draw, resize handles, snap-to-grid, and a real-time code generator supporting 4 component types (Button, Input, Card, Navbar) across 3 output formats.

Gesture-Controlled Snake Game

Python, OpenCV

- Designed a real-time hand-gesture recognition system using MediaPipe Hands + OpenCV achieving approximately 95% landmark detection accuracy at 30 FPS on standard webcam input.
- Reduced gesture-to-action latency to under 80ms by optimizing the frame-processing pipeline, enabling smooth, responsive in-game control without physical input devices.

FotoForFun

ReactJS, HTML, CSS, JavaScript

- Engineered a browser-based photo booth platform supporting real-time webcam access and dynamic image filters.
- Implemented image capture, preview, and download functionalities to deliver a seamless user experience.

Research & Technical Work

Training a **YOLOv11-based** object detection model for **ISDC 2026 (Software Domain)** on a custom-annotated dataset of **7556 images across 5 classes; currently achieving 84.5% mAP@0.5** with ongoing hyperparameter tuning and augmentation experiments.

First-authored a systematic **review paper** comparing CNN and Transformer-based architectures (ResNet, ViT, Swin Transformer) **across 70+ landmark CV papers; submitted to Springer, currently under review.**

Constructing a PPI network from STRING & PSICQUIC datasets using graph clustering (Louvain/MCL) to identify disease-associated gene modules. (ongoing)

Achievements

Certificate of Merit: Ranked among the **Top 10 students twice** in the IT branch.

Code Golf Contest – **3rd Place** (IEEE WIE): Solved algorithmic problems emphasizing logic optimization and minimal code length.

Business Blaze Ideathon – **1st Place**: Proposed **LeadHers**, a platform addressing funding and mentorship gaps.

Leadership & Activities

Vice Chairperson – **IEEE Women in Engineering (WIE), VIT**: Leading chapter initiatives, coordinating technical events and hackathons, and promoting women's participation in technology and engineering.

Event Coordinator – **Switchuation 2.0** (IEEE WIE Hackathon).

Workshop Facilitator – **IEEE Women in Engineering**: Conducted Frontend workshop with a live mini-project.